

Interactive Context-Aware Furniture Recommendation using Mixed Reality

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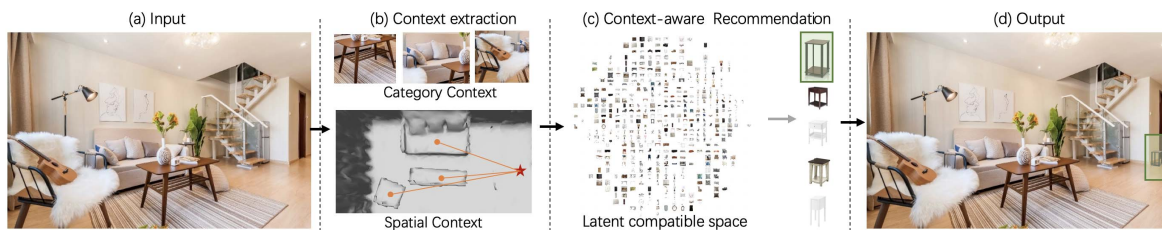


Figure 1: The framework of the proposed system. Given an input scene captured by the camera from the MR device, our approach extracts the contexts and projects them into the latent compatibility space. The recommendation is achieved by finding the best furniture item that has the shortest distance in the latent space. The final recommendation is later augmented on the hologram so that the user can interactively view and manipulate the recommended furniture.

ABSTRACT

We present a Mixed Reality (MR) system, through HoloLens, to provide *context-aware* furniture recommendation in an interactive fashion. Firstly, a ranking-based metric learning method is adopted to represent the furniture compatibility through a latent space. Then, in the recommendation process, a physical scene is captured by the cameras mounted on the MR device, and two types of scene context are analyzed: (1) category context; (2) spatial context. At last, the one with the minimal weighted ranking distance in the latent space is recommended to the user. With MR devices, a user could perceive and manipulate the recommended furniture in real-time. We conduct user study to validate the efficacy of the proposed system.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques; Human-centered computing—Recommendation system—Furniture recommendation

1 INTRODUCTION

Recently, there has been an increasing growth of e-commerce. A variety of products are sold and recommended online. Furniture products also become the upstart of online shopping. Some retailers, *e.g.* IKEA, Wayfair, provide apps or build up systems to recommend and visualize the furniture so that users may have an intuitive perception of how the products look like in the physical world.

User, in most cases, has a very clear goal in terms of the furniture category but lacks intuitive means to facilitate the design choice. Thus we propose an automatic *context-aware* furniture recommendation system in MR environment. We learn a furniture compatibility latent space. The space is learned based on a collected scene dataset, in which the scene is arranged by professional designers. Usually, the furniture have strong relations with each others so that the current furniture items do not affect the desired furniture equally. Thus,

we introduce *category context* and *spatial context*, extracted from the physical world of the user, to weight the obtained compatibility distance.

2 RELATED WORK

In recent years, the evolving technologies of the interactions (*i.e.*, AR and MR) have been reshaping user's expectations and bringing up new user experiences when combined with recommendation systems. For example, we have seen some virtual try-on systems [1]; the system automatically augmented virtual cloth on users, so that users have a more intuitive perception of the effects as if they are trying physical clothes. Similarly, IKEA provides mobile AR apps that allow users to see how new furniture looks in their homes without measuring the physical environment. In academia, Lu *et al.* [4] conducted a study of AR for online shopping, wherein users can select, view, and "try" their favorite products in AR before purchase. Olsson *et al.* [5] presented a study of user experiences with AR in a shopping center and reported mainly positive feedback for mobile AR-supported shopping experiences.

In contrast to the present work, most current AR- and MR-based recommendation systems only use the platform as a visualization tool. The state-of-the-art computer vision methods have been rarely adopted to further the user experiences in the recommendation systems. Leveraging such visual information, in particular the semantics of the environment, could provide a more accurate and intelligent recommendation with improved user experience.

3 SYSTEM FRAMEWORK

Figure 1 shows the framework of the proposed system. A user wears a MR helmet (*e.g.*, HoloLens); through the camera mounted on the helmet, a real scene is captured automatically in (a). The user can specify a desired furniture category using an interface. Then, by analyzing the extracted scene context (b), our system recommends a furniture item and a position to put it accordingly, compatible with the input scene (c). The recommended furniture is automatically placed in the scene, and the user could interactively observe the augmented scene through the MR helmet, as is shown in (d).

Context Extraction. We consider two types of contextual information: category context and spatial context. To extract the category

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Figure 2: Recommendation results. (a) Input scenes. (b) Detected correlated objects. (c) Results. (Top) Recommendations are bookshelf, chair, and coffee table. (Bottom) Recommendations are table, nightstand, and chair.

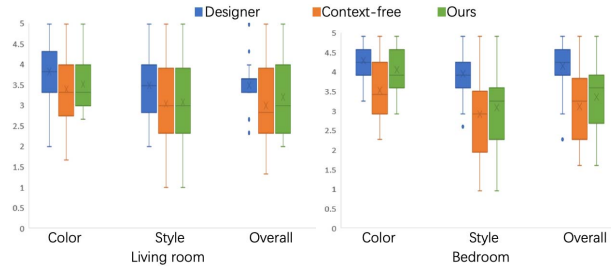


Figure 3: The participants' ratings on the recommendation generated by our approach, designer, and context-free approach. The bottom and top edges of the boxes indicate the 25th and 75th percentiles, respectively. The horizontal lines indicate the median rating. The whiskers extend to the most extreme data points. The circle indicates the outlier rating.

context, we apply a state-of-the-art object detector [2] to detect all furniture items in the scene. For each of the detected object category, we calculate its correlation with the desired furniture category based on a pre-trained word embedding model. To extract the spatial context, our system locates all the furniture items in 3D space by leveraging the 3D reconstruction capability provided by MR devices. The distance between each detected item and the candidate position serves as the spatial context. Both the category context and the spatial context are applied to weight the distance in the latent compatibility space to form the final ranking distance.

Context-aware Recommendation. To estimate the compatibility among furniture items, we adopt recommendation methods to identify and understand compatible relationships across categories [3]. We learn a compatibility space based on a collected furniture compatibility dataset. In such a space, a furniture pair is close if they are compatible with each other. We rank the compatibility of each candidate using the pair-wise distances.

We map all furniture categories and instances into the compatibility space. The extracted contextual information is used to weight the compatibility coefficients to search for the most compatible item. The candidate with the minimum weighted distance is output as the recommendation result. Figure 2 shows the recommendation results.

4 EXPERIMENT

We implement our system using C# in Unity 2017 and deploy it to the HoloLens platform. See qualitative results of a living room and a bedroom scenario in Figure 2. And we conduct a user study to validate the efficacy of the proposed system. We compared the recommendation results by our approach with two alternative approaches; all approaches use the same set of scenes as the input. Specifically, the compared approaches include:

(1) The proposed system; (2) Designer: We recruited a designer

who has about 10-year experiences of interior design. By observing the input scenes, the designer chose one of the items among all candidates to be the most compatible one given the current scene; (3) Context-free approach: This is equivalent to disabling the context extraction component.

All participants were given 18 sets of the input scenes and the recommended items; 6 sets are the recommendations using the proposed system as shown in Figure 2, 6 sets are from the designer's choice, and 6 sets are generated by the context-free approach. The participants were required to score each set in terms of color compatibility, style compatibility, and overall compatibility; the score ranges from 1 to 5.

Participants ratings are shown in Figure 3 using boxplot. We perform a Friedman test on the participants' ratings on all recommendation pairs; the results reveal a significant difference among three approaches using three evaluation metrics (color, style, and overall). For the style metric, there is a significant difference in the bedroom, but not for the living room. Take the overall ratings as an example; there is a significant difference among three approaches for living room ($\chi^2 = 6.63, p < .05, df = 2$) at the $\alpha = 0.05$ significance level, and similarly, ($\chi^2 = 28.49, p < .05, df = 2$) for bedroom.

We further conduct post-hoc tests on every two approaches using t-test at the $\alpha = 0.05$ significance level. For the bedroom, the results show a significant difference ($t(23) = -3.55, p < .05$), and the proposed system (Md = 3.67) obtained higher ratings than the ones recommended by the context-free approaches (Md = 3.33). We also observe a significant difference between the results recommended by the designer (Md = 4.33) and by the context-free approaches (Md = 3.33) ($t(23) = 1.59, p < .05$). The ratings of the recommended items by the designer approach (Md = 4.33) and the proposed system (Md = 3.67) have no significance.

For the living room, the median overall ratings of the recommendation generated by the proposed system, the designer, and the context-free approach are 3, 3.67, and 2.83, respectively. We do not observe significant differences between every pair of the two approaches; a possible reason is that most pieces of furniture in the living room have similar attributes, e.g., color, style. The t-test results on the color and style ratings also show no significant difference.

5 CONCLUSION

As an early attempt to apply the contextual information to the furniture recommendation in a MR environment, we consider category and spatial context in the recommendation. There are other contextual information which may also help to improve the recommendation quality, e.g., the room size, user's attributes and so on. In the future, exploring more contextual information is an interesting direction. Of course, our pipeline can also be adopted to complete other recommendation tasks. For example, our approach may recommend decoration to match the style of a real scene.

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